

Automated Student Attendance Notifier

Project Overview

Objective

➔ To automate student attendance tracking by comparing uploaded image filenames with registered student names, updating attendance records, and notifying absentees via email.

Team Members

- **Abhyendra Singh**
- **Harsh Sahal**
- **Pawan Singh Chouhan**

Mentor

- **Mr. Ashutosh Singhal**
Regex Software Services

Technologies Used

Technology		Purpose
AWS S3	->	Store student images (registration & attendance)
AWS DynamoDB	->	Store attendance records
Amazon SES	->	Send email alerts to absent students
AWS Lambda	->	Automate logic: name matching, DB updates, email triggering

System Workflow

End-to-End Flow

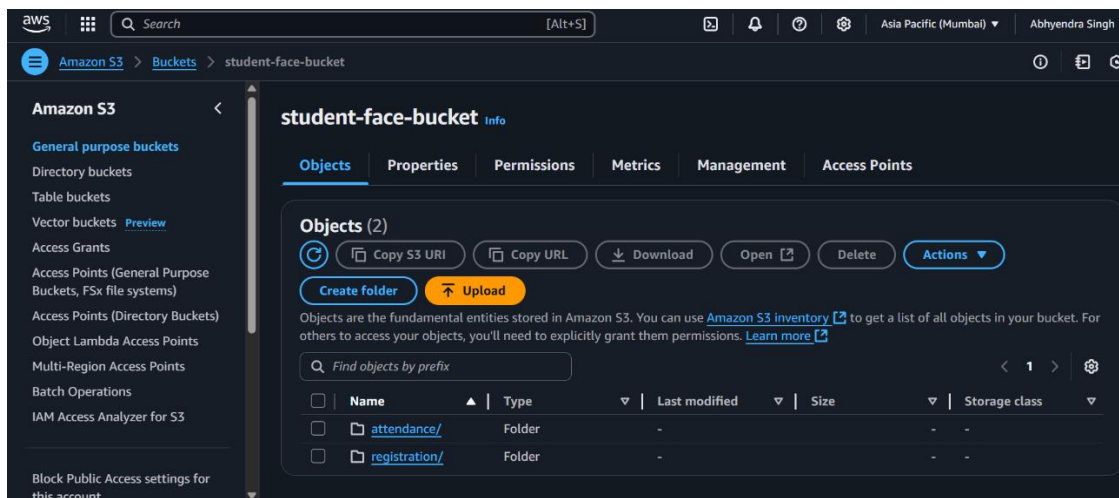
1. Student registration images uploaded to S3 → registration/
2. Daily attendance images uploaded to S3 → attendance/

3. S3 triggers Lambda function
4. Lambda compares filenames with registered names
5. DynamoDB updated with Present/Absent status
6. SES sends email to absent students

Implementation Steps with Screenshots

1 S3 Bucket Structure

 -> *student-face-bucket showing registration/ and attendance/ folders*

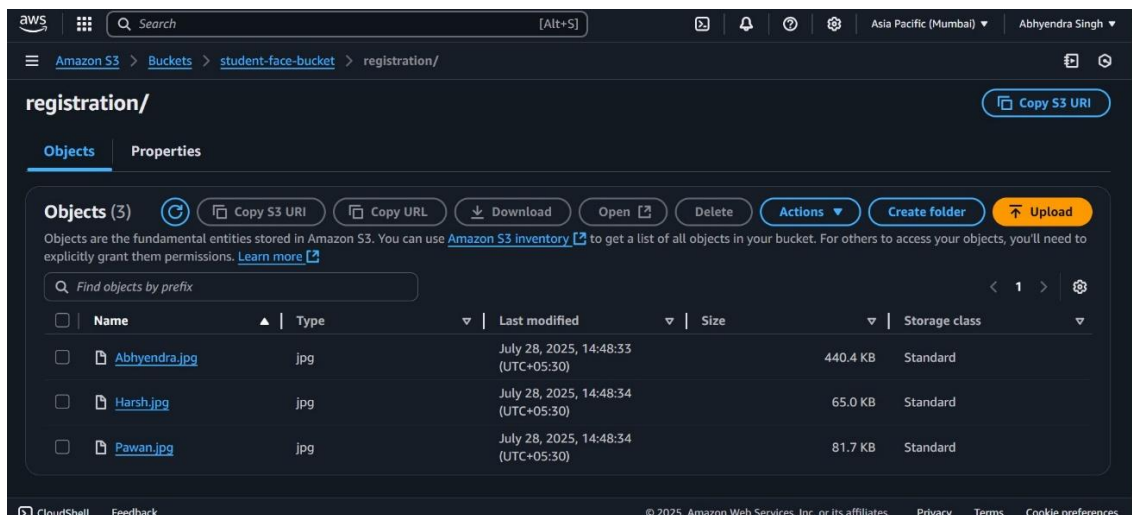


 **Explanation:**

- registration/: Stores student images named by their names
- attendance/: Stores daily attendance images
- Matching is done by comparing filenames

2 Uploading Registration Images

 -> *Upload confirmation of Abhyendra.jpg, Pawan.jpg, Harsh.jpg to registration/*

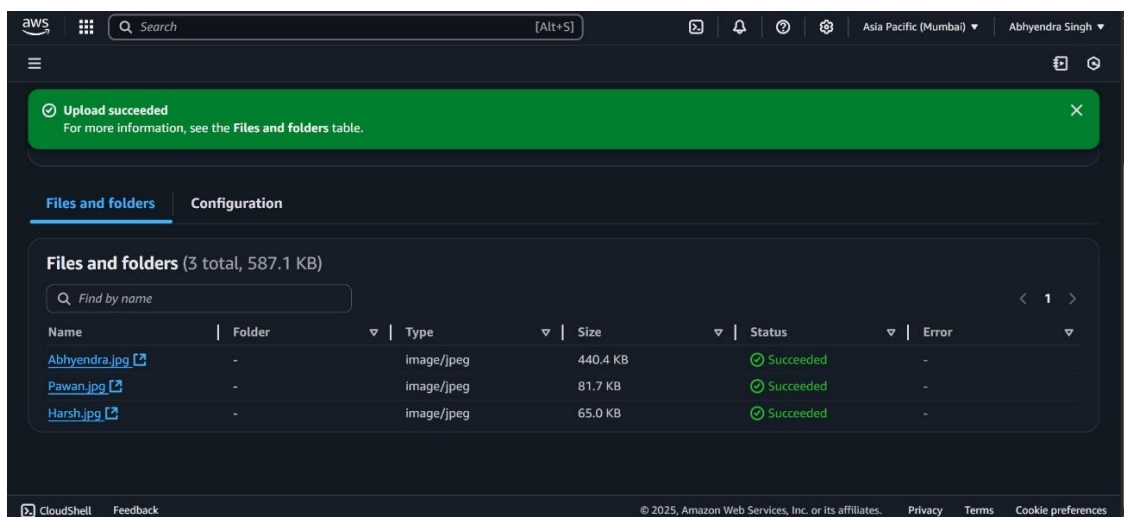


Explanation:

- These images represent registered students
- Stored in registration/ folder for reference

3 Attendance Images Uploaded

-> Upload confirmation of same three images to attendance/



Explanation:

- System compares names from filenames
- All three matched → marked Present

4 DynamoDB Table — All Present

📷 -> *StudentAttendance table showing all students as Present*

The screenshot shows the AWS DynamoDB console interface. On the left is a navigation menu with options like Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main area displays the 'StudentAttendance' table. A green notification bar at the top indicates a successful scan: 'Completed · Items returned: 3 · Items scanned: 3 · Efficiency: 100% · RCUs consumed: 2'. Below this, a table titled 'Table: StudentAttendance - Items returned (3)' shows the scan results. The table has columns for StudentName (String), Date (String), and Status. The data rows are:

StudentName (String)	Date (String)	Status
Pawan	2025-07-28	Present
Abhyendra	2025-07-28	Present
Harsh	2025-07-28	Present

📝 *Explanation:*

- Lambda matched all names
- DynamoDB updated with Status = Present

5 New Attendance Upload — Only Abhyendra & Pawan

📷 -> *Upload confirmation of Abhyendra.jpg and Pawan.jpg*

The screenshot shows the AWS S3 console interface. A green notification bar at the top indicates a successful upload: 'Upload succeeded · For more information, see the Files and folders table.' Below this, the 'Files and folders' tab is selected, showing a list of files. The table has columns for Name, Folder, Type, Size, Status, and Error. The data rows are:

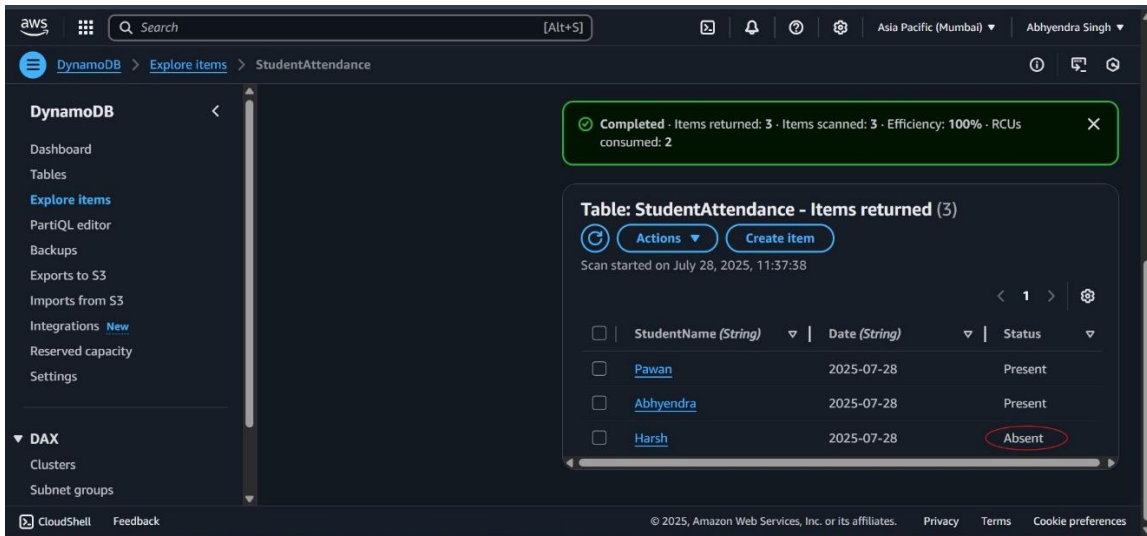
Name	Folder	Type	Size	Status	Error
Pawan.jpg	-	image/jpeg	81.7 KB	Succeeded	-
Abhyendra.jpg	-	image/jpeg	440.4 KB	Succeeded	-

📝 *Explanation:*

- Harsh's image not uploaded today
- Lambda marks Harsh as Absent

6 DynamoDB Table — Harsh Absent

📷 -> Updated table showing Harsh marked Absent



The screenshot shows the AWS DynamoDB console interface. On the left, the navigation menu includes options like Dashboard, Tables, Explore Items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main area displays the 'StudentAttendance' table. A green notification banner at the top right indicates a successful scan: 'Completed - Items returned: 3 - Items scanned: 3 - Efficiency: 100% - RCUs consumed: 2'. Below this, the table 'Table: StudentAttendance - Items returned (3)' is shown with a 'Scan started on July 28, 2025, 11:37:38'. The table has three columns: StudentName (String), Date (String), and Status. The data rows are: Pawan (2025-07-28, Present), Abhyendra (2025-07-28, Present), and Harsh (2025-07-28, Absent). The word 'Absent' for Harsh is circled in red. The bottom of the console shows the footer with copyright information and links to Privacy, Terms, and Cookie preferences.

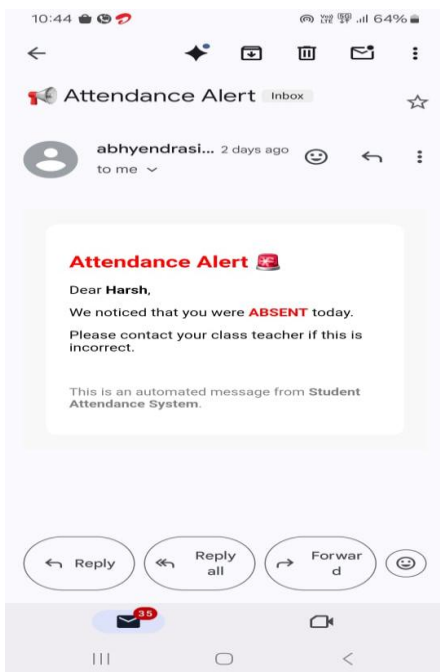
StudentName (String)	Date (String)	Status
Pawan	2025-07-28	Present
Abhyendra	2025-07-28	Present
Harsh	2025-07-28	Absent

📄 *Explanation:*

- Name not found in today's attendance folder
- Marked Absent in DynamoDB

7 Email Alert — Harsh Absent

📷 -> Email sent to Harsh



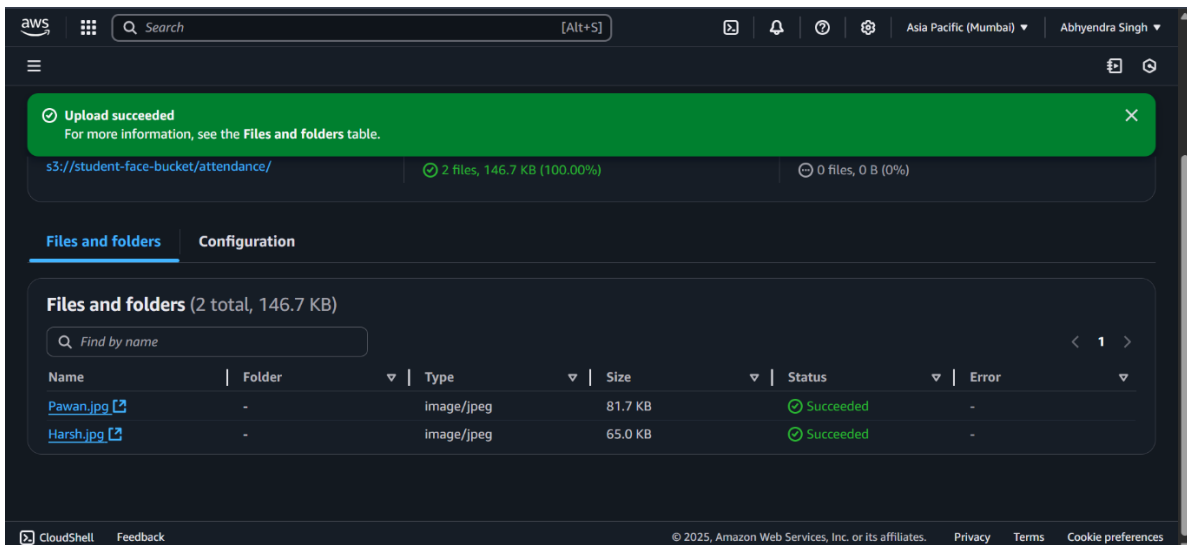
The screenshot shows an email interface on a mobile device. The email is titled 'Attendance Alert' and is from 'abhyendrasi... 2 days ago to me'. The body of the email contains the following text: 'Attendance Alert', 'Dear Harsh,', 'We noticed that you were **ABSENT** today.', 'Please contact your class teacher if this is incorrect.', and 'This is an automated message from Student Attendance System.' At the bottom of the email, there are buttons for 'Reply', 'Reply all', and 'Forward'. The bottom of the screen shows the mobile home indicator bar.

 **Explanation:**

- Triggered by Lambda
- Sent via Amazon SES
- Message advises student to contact class teacher

8 New Attendance Upload — Harsh & Pawan Only

 -> Upload confirmation of Harsh.jpg and Pawan.jpg



The screenshot shows the AWS S3 console interface. At the top, a green notification bar states "Upload succeeded" for the path `s3://student-face-bucket/attendance/`, indicating that 2 files (146.7 KB) were uploaded successfully. Below this, the "Files and folders" tab is active, displaying a table with the following data:

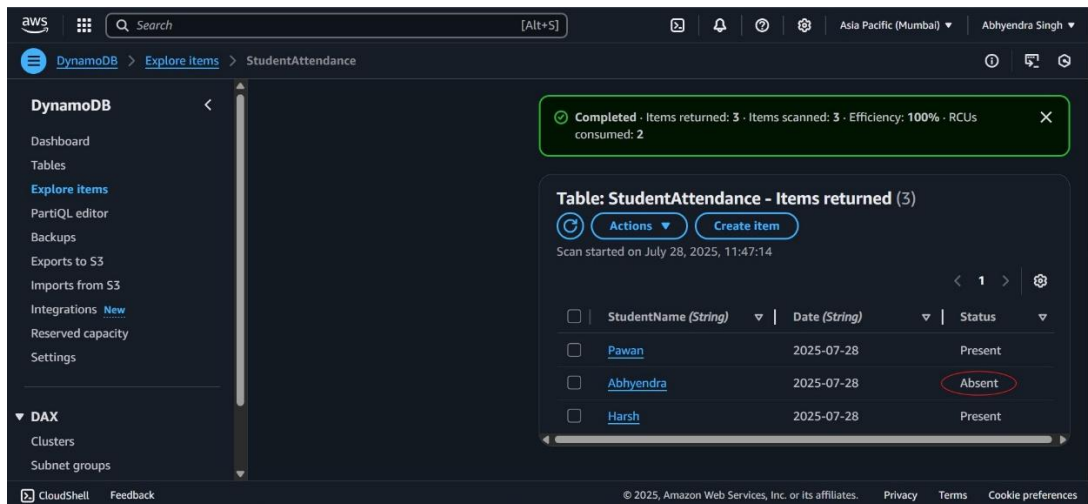
Name	Folder	Type	Size	Status	Error
Pawan.jpg	-	image/jpeg	81.7 KB	Succeeded	-
Harsh.jpg	-	image/jpeg	65.0 KB	Succeeded	-

 **Explanation:**

- Abhyendra's image missing today
- Lambda marks Abhyendra as Absent

9 DynamoDB Table — Abhyendra Absent

 -> Updated table showing Abhyendra marked Absent

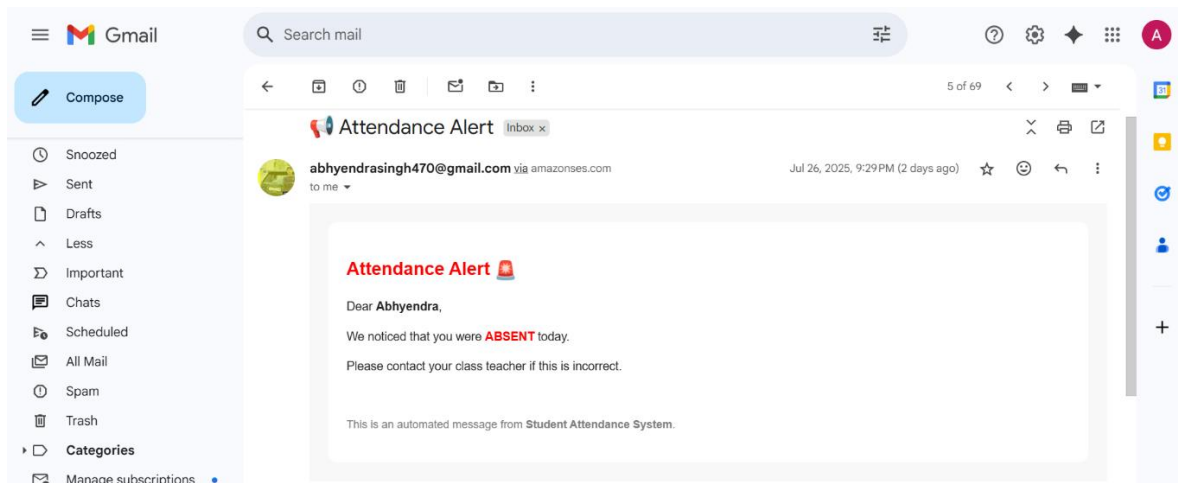


Explanation:

- Name not found in attendance folder
- Marked Absent in DynamoDB

10 Email Alert — Abhyendra Absent

-> Email sent to Abhyendra



Explanation:

- Automated alert via SES
- Message includes absence notification and contact advice

Conclusion

This system provides:

-  Simple and effective attendance tracking using filename matching
-  Secure image storage via AWS S3
-  Real-time updates in DynamoDB
-  Automated alerts for absentees via SES
-  Fully serverless and event-driven architecture using AWS Lambda